

08 - Design and Implementation of API Publication tool and schema mapping in E-invoicing API Ecosystem

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Allowed groups: 6 groups

Project background & goals

Project Background

E-invoicing has rapidly shifted from an optional efficiency improvement to a regulatory obligation across much of the world. Governments are mandating structured electronic invoices to close VAT gaps, improve tax compliance, and standardize cross-border trade — with frameworks such as PEPPOL, EN 16931, and country-specific mandates (Italy's SDI, India's IRP, Saudi ZATCA, France's upcoming reform) reshaping how supply chain enterprises issue, exchange, and archive invoices.

Despite this push toward standardization, the technical reality inside enterprises remains fragmented. Supply chain organizations typically operate a patchwork of legacy ERP systems, custom integrations, and third-party connectors, each exposing its own APIs for invoice creation, validation, transmission, and archiving. These APIs are rarely discoverable beyond the team that built them, often poorly documented, and inconsistent in their formats, authentication mechanisms, and compliance posture. As a result:

1. Enterprises duplicate effort by re-building integrations that already exist elsewhere in the organization or industry.
2. Onboarding new e-invoicing requirements (new countries, new standards) is slow and costly.
3. There is no systematic way to compose existing technical APIs into higher-level business processes that match how enterprises actually operate.
4. Discovery of suitable APIs depends on tribal knowledge rather than tooling.

What is missing is a structured e-invoicing API ecosystem — a curated repository of domain-specific APIs, paired with tools that allow enterprises to publish, discover, and compose those APIs reliably. The broader project introduces such an ecosystem, consisting of four cooperating components: a central API Repository, a Composition Service (BP-Builder) that turns technical APIs into enterprise-ready business APIs and processes, an AI-powered Discovery Service, and an API Publisher through which enterprises contribute their internal APIs to the repository.

This work focuses on the API Publisher tool — the entry point of the ecosystem and the quality gate that determines whether the repository becomes a trustworthy, composable foundation or a dumping ground of inconsistent APIs.

Project Goals

The overall goal of this work is to design and implement an API Publisher tool that enables supply chain enterprises to contribute their internal e-invoicing APIs to a shared repository in a controlled, standards-compliant, and metadata-rich manner, so that those APIs become readily usable by the Discovery and Composition services.

Specifically, the project aims to:

1. Enable enterprise contribution of internal e-invoicing APIs through a developer-friendly tool integrated into the unified ecosystem UI.
2. Enforce a multi-stage validation pipeline at publish time, covering OpenAPI/Swagger specification correctness, e-invoicing domain compliance, and security and authentication metadata.
3. Capture rich, structured metadata about each API — including name, endpoint URL, protocol (REST or SOAP), input and output formats, and authentication method — sufficient to drive downstream AI-based discovery and automated composition.

4. Support both REST and SOAP APIs, reflecting the realistic mix of modern and legacy services in supply chain environments.
5. Provide a seamless user experience by exposing the Publisher alongside the Discovery and Composition services within a single interface, so enterprises perceive one coherent platform rather than separate tools.
6. Ensure repository quality by rejecting non-compliant or insufficiently described APIs at the point of entry, protecting downstream services from unreliable inputs.

Project scope

This section defines what the API Publisher tool will and will not deliver, clarifying its boundaries with respect to the other components of the e-invoicing API ecosystem. The project covers the design and implementation of the API Publisher tool as a developer-oriented service within the unified ecosystem UI, including the following capabilities:

1. Publisher User Interface

A web-based interface, integrated into the same platform as the Discovery and Composition services, through which enterprise developers submit their internal e-invoicing APIs for publication to the repository.

2. API Submission Workflow

A guided flow that allows publishers to upload an API specification (OpenAPI/Swagger for REST, WSDL for SOAP) and enter the required business and technical metadata, including:

API name

API endpoint URL

Protocol type (REST or SOAP)

Input format

Output format

Authentication method

Additional descriptive metadata required by downstream services

3. Validation Pipeline

A multi-stage validation engine executed at publish time, consisting of:

Specification validation — syntactic and structural correctness of the OpenAPI/Swagger (or WSDL) document.

E-invoicing domain compliance validation — verification against selected e-invoicing standards relevant to supply chain enterprises.

Security and authentication metadata validation — verification that the API declares a recognized and adequately described authentication mechanism.

4. Metadata Persistence

Storage of validated APIs and their associated metadata into the central API Repository in a format consumable by the Discovery Service and the Composition Service.

5. Publisher Feedback

Clear, actionable feedback to the publisher when validation fails, indicating which stage failed and why, so the issue can be corrected and the API resubmitted.

6. Integration with the Ecosystem

Functional integration with the API Repository so that published APIs become immediately available for discovery and composition by the other services.

7. Addition of more schema formats (PDF to XML etc. details will be shared later) to the already established schema mapping framework for APIs

Project requirements

Functional Requirements

FR-1. Publisher Authentication and Authorization

The system shall allow enterprise users to authenticate before accessing the Publisher tool. Only authenticated users belonging to a registered enterprise shall be permitted to publish APIs. The system shall enforce role-based access so that only authorized users within an enterprise can submit, modify, or withdraw APIs.

FR-2. API Specification Upload

The system shall allow the publisher to upload an API specification document. Supported formats shall include OpenAPI/Swagger (for REST APIs) and WSDL (for SOAP APIs). The system shall accept both file uploads and direct URL references to a hosted specification.

FR-3. Metadata Capture Form

The system shall provide a structured form through which the publisher enters business and technical metadata for the API, including at minimum:

API name

API endpoint URL

Protocol type (REST or SOAP)

Input format

Output format

Authentication method (e.g., OAuth 2.0, API key, Basic, mTLS)

Brief functional description of the API

E-invoicing capability category (e.g., invoice creation, validation, transmission, archiving)

FR-4. Specification Validation

The system shall validate the uploaded specification for syntactic and structural correctness against the relevant standard (OpenAPI for REST, WSDL for SOAP). Invalid specifications shall be rejected with a detailed error report.

FR-5. E-Invoicing Domain Compliance Validation

The system shall validate the submitted API against the configured e-invoicing domain rules to ensure that its declared inputs, outputs, and operations are consistent with recognized e-invoicing standards. APIs that fail compliance checks shall be rejected with a domain-specific error report.

FR-6. Security and Authentication Metadata Validation

The system shall verify that the API declares an acceptable authentication mechanism and that the security-related metadata in the specification is complete and internally consistent. APIs that do not meet the security baseline shall be rejected.

FR-7. Validation Feedback

The system shall present validation results to the publisher in a structured, human-readable form, clearly indicating which validation stage passed or failed and the reason for each failure, so the publisher can correct and resubmit.

FR-8. Publication to the Repository

Upon successful completion of all validation stages, the system shall persist the API specification and its associated metadata into the central API Repository in a format consumable by the Discovery Service and the Composition Service.

FR-9. Draft and Resubmission Support

The system shall allow publishers to save in-progress submissions as drafts and resubmit corrected APIs after validation failures without re-entering previously provided information.

FR-10. Listing of Published APIs

The system shall allow publishers to view the list of APIs their enterprise has already published, including each API's current status (e.g., published, rejected, draft).

FR-11. Update and Withdrawal of Published APIs

The system shall allow publishers to update the metadata or specification of a previously published API (subject to revalidation) and to withdraw an API from the repository.

FR-12. Integration with the Unified UI

The system shall be accessible as one of the services within the same unified user interface that hosts the Discovery Service and the Composition Service, providing a consistent navigation and user experience across the ecosystem.

FR-13. Mapping one API to another API based on their schemas, and adding more format mappings to the framework for API schema mapping.

Required knowledge and skills

Domain Knowledge

1. Fundamentals of e-invoicing — understanding what an electronic invoice is, why it is regulated, and the role it plays in supply chain and tax compliance processes.
2. Awareness of e-invoicing standards — familiarity with at least one widely used framework such as PEPPOL BIS Billing 3.0, EN 16931, or UBL 2.1, including their typical document structures and validation rules.
3. Understanding of enterprise integration concerns — appreciation of how supply chain enterprises operate heterogeneous, often legacy, systems and why standardized API publication matters in that context.

Technical Knowledge

1. API design and architecture — solid understanding of REST principles, HTTP semantics, and resource modeling, as well as a working knowledge of SOAP-based services and the role of WSDL.
2. API specification languages — practical familiarity with OpenAPI/Swagger (for REST) and WSDL (for SOAP), including how to read, write, and programmatically validate such documents.
3. Data formats — comfort with JSON and XML, and awareness of domain-specific XML dialects used in e-invoicing (e.g., UBL).
4. Authentication and API security — understanding of common API authentication mechanisms (OAuth 2.0, API keys, Basic authentication, mTLS) and basic principles of secure data handling in transit and at rest.
5. Web application development — ability to build a web-based tool with a modern front-end framework and a back-end service, including form handling, file upload, and integration with downstream services.
6. Software engineering practices — version control (Git), modular design, unit and integration testing, and documentation of code and APIs.

Tools and Technologies

1. Working knowledge of at least one back-end language and framework suitable for building validation services (e.g., Java/Spring, Python/FastAPI, Node.js/Express, or equivalent).
 2. Working knowledge of at least one front-end framework (e.g., React, Angular, or Vue) for implementing the Publisher UI within the unified ecosystem interface.
 3. Familiarity with validation libraries and tooling for OpenAPI/Swagger and XML/WSDL.
- Basic familiarity with containerization (e.g., Docker) and with common development tools such as Postman or equivalent for API testing.

Expected outcomes and deliverables

Expected Outcomes

By the end of the project, the following outcomes are expected:

1. A working API Publisher tool that enables supply chain enterprises to publish their internal e-invoicing APIs to the central repository through a developer-friendly interface integrated into the unified ecosystem UI.
2. A functioning multi-stage validation pipeline that systematically checks each submitted API for specification

correctness, e-invoicing domain compliance, and security and authentication metadata adequacy before admitting it to the repository.

3. A higher-quality API Repository, populated only with APIs that meet defined structural, domain, and security criteria, and therefore suitable as a reliable input to the Discovery Service and the Composition Service.

4. Demonstrated end-to-end integration within the ecosystem, showing that an API published through the Publisher can subsequently be discovered by the Discovery Service and used by the Composition Service to build a business API or business process.

5. A reusable design and reference implementation that can be extended to additional e-invoicing standards, protocols, or validation rules in future work, and that can inform similar API publishing tools in other regulated domains.

Deliverables

The following concrete deliverables are expected at the conclusion of the project:

1. Source Code

The complete source code of the API Publisher tool, including:

2. Front-end implementation of the Publisher UI within the unified ecosystem interface.

3. Back-end services implementing the validation pipeline and repository integration.

4. Configuration files, build scripts, and dependency definitions required to run the system.

The code shall be maintained in a version-controlled repository with a clean commit history.

5. Validation Modules

The implementations of the three validation stages — specification validation, e-invoicing domain compliance validation, and security/authentication metadata validation — delivered as modular components with clearly defined inputs, outputs, and extension points.

Technical Documentation

A document describing:

6. System architecture and how the Publisher fits within the broader ecosystem.

Detailed design of the validation pipeline, including the standards and rules applied at each stage.

Data model and metadata schema used to describe published APIs in the repository.

Interfaces exposed to and consumed from the API Repository, Discovery Service, and Composition Service.

7. User Guide

A guide aimed at enterprise developers using the Publisher, covering:

How to access the Publisher within the unified UI.

How to prepare an API specification for submission.

How to fill in the required metadata.

How to interpret validation results and resolve common errors.

How to manage previously published APIs (update, withdraw, view status).

8. Developer and Deployment Documentation

Instructions covering how to build, configure, and deploy the Publisher in a development or test environment, including any prerequisites, environment variables, and integration steps with the API Repository.

9. Test Suite and Test Report

A suite of automated tests covering the validation pipeline and key user workflows, together with a short report summarizing test coverage and results.

Sample APIs and Demonstration Scenario

A small collection of representative sample e-invoicing APIs (both REST and SOAP) used to demonstrate and validate the Publisher, together with a documented end-to-end scenario showing publication, discovery, and composition of at least one API.

10. Final Project Report

A written report covering the project background, problem statement, requirements, design decisions, implementation, evaluation, limitations, and possible future work.

Disciplines related to the project

Software Development;Web Application Development;Artificial Intelligence (Machine/Deep Learning, NLP);Generative AI (GenAI);Big data Analytics and Visualization;

Resources provided

will be shared later